

전자회로 이론(공)(반) (SM304-01)

학기	2026년 1학기
담당교수	이정협
이수구분	심화필수
학점	3.0
강의시간	화14:30-16:00(E7 - 235), 목14:30-16:00(E7 - 235)
강의실	E3 509 (이정협 교수)
소속	반도체공학과

교과개요

This course introduces basic physical electronics, electronic devices, and device characteristics. The devices include diodes, metal oxide semiconductor field-effect transistors (MOSFET), bipolar junction transistors (BJT), and operational amplifiers (Op-Amp). The course provides the fundamentals and basic skills required to analyze (DC, AC and small-signal analysis) and design circuits containing diodes, MOSFETs, BJTs, and Op-amps. The course begins with the explanation on the device fundamentals prior to building circuits. The device becomes the building block of analog/digital circuits. Then, we cover basis of diode models to get familiar with the pn junction devices. Before looking into complicate operational amplifier, the course deals with the design of amplifiers using BJT. Then, we discuss the fundamentals of MOS amplifiers. At later stage of the course, we will talk about more complex amplifier designs, such as differential amplifiers.

교과목표

The aim of the course is for: - Understanding the fundamental characteristics of semiconductor devices and diodes. - Understanding the concepts of bipolar junction transistors and metal oxide semiconductor field-effect transistors. - Understanding the concepts of operational amplifiers which become the basic building blocks of analog circuits. - Understanding the behavior of amplifiers in frequency domain.

학습윤리

No tolerance for cheating